Multi-task Learning of Negation and Speculation for Targeted Sentiment Classification

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What is targeted sentiment classification?

System Input:

June found the **Welsh fruit cake** to be tasteless but the **chocolate cake** to be great.

System Output:

1. Welsh fruit cake – Negative
2. Chocolate cake – Positive

1. What is the effect of negation and speculation on targeted sentiment classification?

2. How can we make models more robust to negation and speculation?

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Challenge Dataset:

Type	Sentence	Sentiment towards *sushi*
Original	This is good, inexpensive sushi.	Positive
Negated	This is **not** good, inexpensive sushi.	Negative
Speculative	**I'm not sure if** this is good, inexpensive sushi.	Neutral

Challenge Dataset

Dataset	Train	Dev	Test	Inter-Annotator Agreement (IAA)
Laptop	2,044	256	634	0.67
Laptop_{Neg}	-	181	470	0.70
Laptop_{Spec}	-	142	220	0.64
Restaurant	3,896	414	2,288	0.71
Restaurant_{Neg}	-	274	1,013	0.66
Restaurant_{Spec}	-	200	451	0.66

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How can we make models more robust to negation and speculation?

1. Transfer Learning (**TL**)
2. Multi-task learning (**MTL**)

Task Setup

Sequence labelling – BIOUL with sentiment labels

Sentiment labels – Positive, negative, and neutral.

O	O	O	B_{neg}	I_{neg}	L_{neg}	O	O	O	O	O	B_{pos}	L_{pos}	O	O	O	O
June	found	the	**Welsh**	**fruit**	**cake**	to	be	tasteless	but	the	**chocolate**	**cake**	to	be	great	.

1. Welsh fruit cake – Negative
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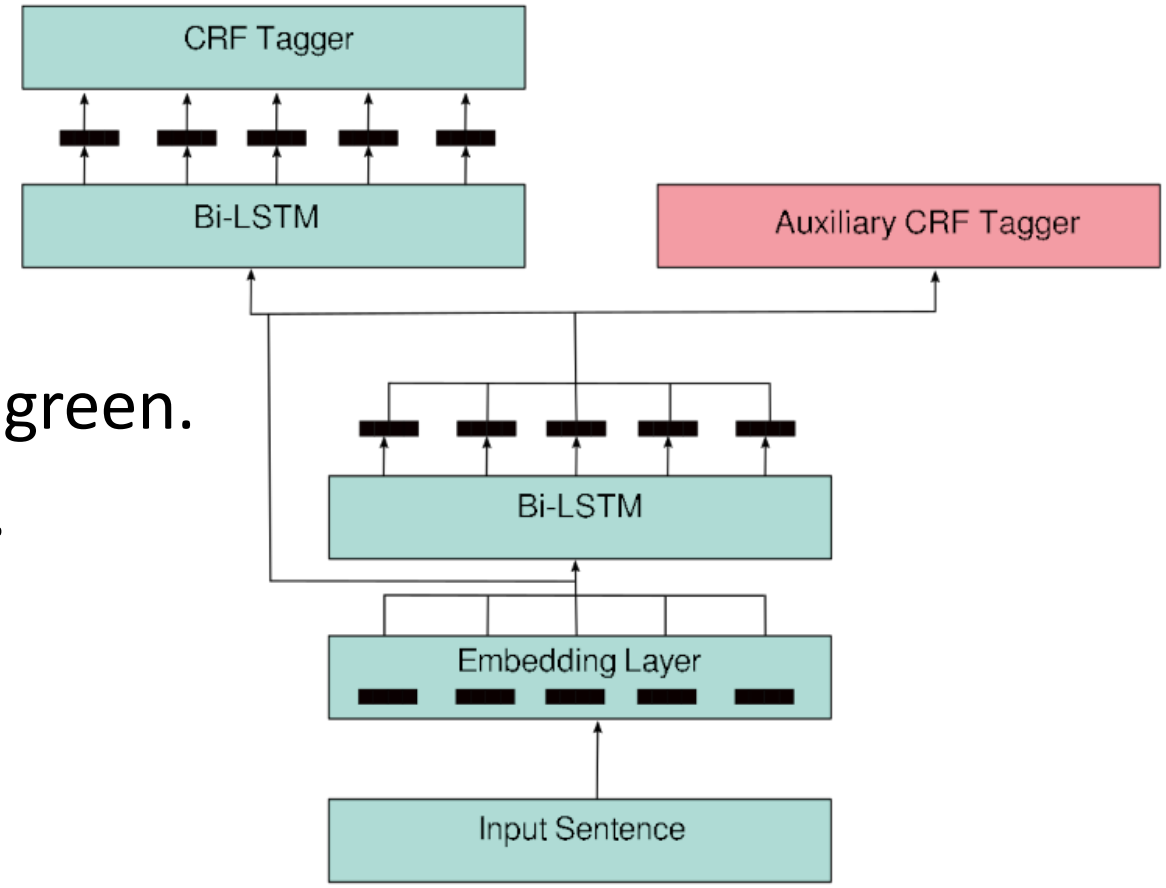
Baseline Model



- Single Task Learning (**STL**) layers are green.
- Pink layer is for the MTL model only.
- Embedding layer is always frozen.

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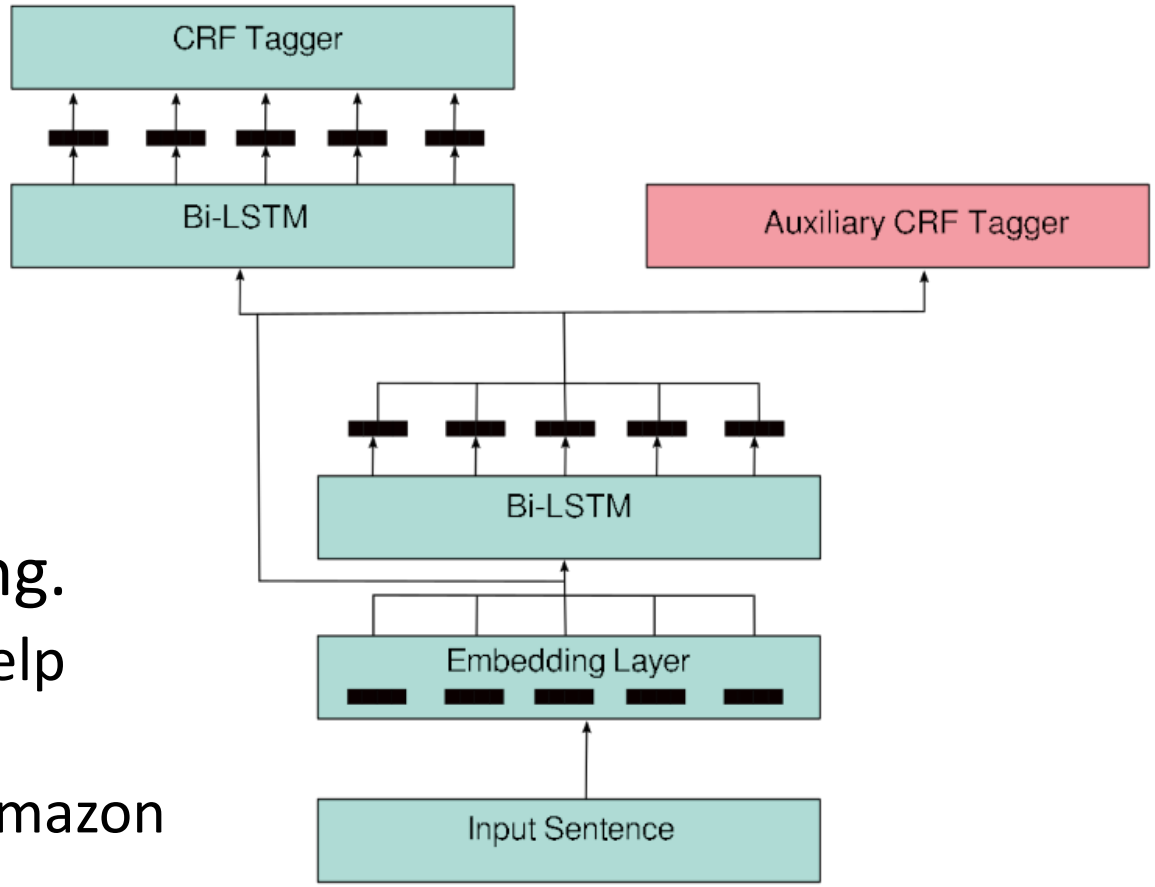
Transfer Learning (TL)

- GloVe – Non Transfer Learning
  - Pre-trained on English Wikipedia and Gigaword
- Transformer ELMo – Transfer Learning.
  - Restaurant datasets – pre-trained on Yelp reviews
  - For Laptop datasets – pre-trained on Amazon electronic reviews



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Multi-task Learning (MTL)
- Auxiliary tasks

- Negation (**Neg**) – Conan Doyle (**CD**) book dataset
- Negation (**Neg**) – **SFU** review dataset
- Speculation (**Spec**)
- Universal Part Of Speech (**UPOS**) tagging
- Dependency Relation (**DR**) prediction
- Lexical Analysis (**LEX**)

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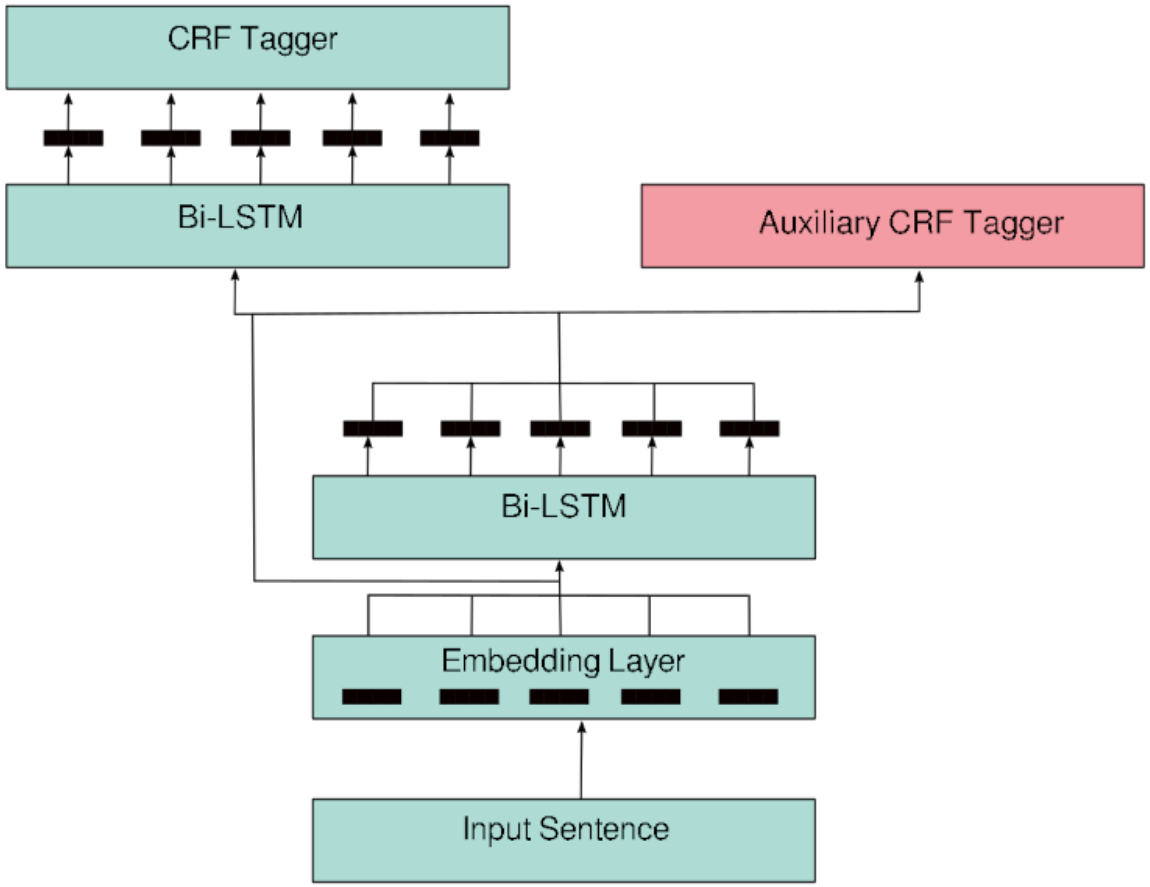

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Metrics

- Sentiment – The accuracy score for the sentiment classification for all targets that have been correctly extracted.
- Extraction – F1 score of target extraction.
- Targeted – F1 score of correctly extracting the target and the polarity towards that target.

Baseline Results – Metric Targeted.

1. Transfer Learning improves results on average by 7.21 percentage points.
2. MTL is always equivalent in performance to it's STL equivalent.

		Laptop		Restaurant	
	Auxiliary	GloVe	TL	GloVe	TL
MTL	NEG_{CD}	54.65	62.89	65.06	71.04
	DR	53.67	62.29	**66.06**	71.45
	LEX	**54.85**	62.55	65.89	**71.77**
	NEG_{SFU}	53.73	62.61	65.82	71.63
	SPEC	51.65	62.03	65.16	71.51
	UPOS	54.17	62.35	65.73	70.38
STL		54.37	**63.70**	65.60	70.68

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Negation Results – Metric Targeted.

1. On average all models perform **24 percentage points worse** compared to the original test set.
2. The best performing MTL GloVe model is statistically significantly better than it's STL equivalent.
3. Transfer learning improves results on average by 9.55 percentage points.

		$\text{Laptop}_{\text{Neg}}$		$\text{Restaurant}_{\text{Neg}}$	
	Auxiliary	GloVe	TL	GloVe	TL
MTL	NEG_{CD}	32.28	**39.95**	43.28	50.40
	DR	29.30	38.08	40.95	**51.92**
	LEX	30.47	38.35	39.19	51.15
	NEG_{SFU}	**33.96**	39.18	**45.22**	50.75
	SPEC	31.36	37.59	39.85	51.49
	UPOS	30.36	38.23	41.43	49.45
STL		29.33	38.14	40.38	50.68

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Speculation Results – Metric Targeted.

1. On average all models perform **25 percentage points worse** compared to the original test set.
2. Transfer learning improves results on average by 3.65 percentage points.

		$Laptop_{Spec}$		$Restaurant_{Spec}$	
	Auxiliary	GloVe	TL	GloVe	TL
MTL	NEG_{CD}	25.67	28.36	47.14	52.39
	DR	26.39	28.09	**48.94**	53.69
	LEX	**28.59**	28.56	48.07	**55.15**
	NEG_{SFU}	27.33	28.33	47.90	53.25
	SPEC	26.95	28.54	48.41	52.34
	UPOS	27.31	28.37	47.94	52.55
STL		26.01	**28.72**	48.35	52.94

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Findings

- For both negation and speculation results the majority of error came from sentiment classification error rather than extraction.
- Auxiliary tasks improved results on different metrics and different challenge datasets.
- MTL and transfer learning do no always complement each other.

Contributions

- Quantify the effects of negation and speculation for Targeted Sentiment Analysis. Finding that all models perform on average 24 and 25 percentage points worse on the negation and speculation datasets.
- Created two English challenge datasets annotated for negation and speculation.
- Find both transfer learning and multi-task learning can create more robust models to the effects of negation and speculation.

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Thanks!

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Jeremy Barnes – University of Oslo

Code / challenge dataset / trained models:

https://bit.ly/33yyRLI

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